

Data Dictionary — Sentinel-2 Vegetation Indices

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

Source: Sentinel-2 SR Harmonized (COPERNICUS/S2_SR_HARMONIZED) via Google Earth Engine
Native resolution: 10–20 m · **Revisit:** ~5 days (combined S2A + S2B)
Period: 2019-10-01 → 2024-07-31 (5 wheat seasons) · **Parcels:** 1,500
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1. Overview

This dataset provides Sentinel-2 vegetation indices (NDVI, EVI, NDWI) extracted at the parcel level for 1,500 winter wheat fields across Centre-Val de Loire. Three temporal granularities are delivered to support different machine-learning architectures:

- **15-day composites** (118 windows, 5 statistics): classical ML baseline
- **7-day composites** (253 windows, 5 statistics): sequence models (LSTM)
- **Raw scene-by-scene values** (5 yearly files): maximum flexibility

2. File inventory

Filename	Approx size	Rows (incl. NaN)	Window
sentinel2_composites_15days_cv1.csv	42 MB	~177,000	15 days
sentinel2_composites_7days_cv1.csv	~85 MB	~380,000	7 days
sentinel2_raw_2020.csv	~40 MB	~1,120,000	per scene
sentinel2_raw_2021.csv	~40 MB	~1,150,000	per scene
sentinel2_raw_2022.csv	~40 MB	~1,150,000	per scene
sentinel2_raw_2023.csv	~40 MB	~1,150,000	per scene
sentinel2_raw_2024.csv	~40 MB	~1,120,000	per scene

3. Column dictionary — Composite files (15j and 7j)

KEY identifier **META** metadata **FEATURE** vegetation index statistic

Column	Type	Unit	Description
parcelle_uid KEY	str	—	Unique observation ID (year_IDPARCEL). Primary join key with weather, soil, target.
year KEY	int	year	Harvest year (2020–2024).
window_start META	date	YYYY-MM-DD	Start date of the composite window.
window_days META	int	days	Window length (7 or 15).
n_images META	int	—	Number of valid scenes aggregated in the window (after cloud masking).
NDVI_median FEATURE	float	[-1, 1]	Median NDVI across scenes in the window (robust central value).
NDVI_mean FEATURE	float	[-1, 1]	Mean NDVI across scenes.
NDVI_max FEATURE	float	[-1, 1]	Maximum NDVI in the window (peak vigour reference).
NDVI_min FEATURE	float	[-1, 1]	Minimum NDVI in the window.
NDVI_stdDev FEATURE	float	≥ 0	Standard deviation of NDVI in the window (temporal variability).
EVI_{median, mean, max, min, stdDev} FEATURE	float	[-1, 1]*	Same five statistics for Enhanced Vegetation Index.

NDWI_{median, mean, max, min, stdDev}	FEATURE	float	[-1, 1]	Same five statistics for Normalized Difference Water Index.
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* EVI can produce extreme values due to its formula (~0,12% of rows). Filter $|EVI| < 1,5$ before modelling.

4. Column dictionary — Raw files (one per year)

Column	Type	Unit	Description
parcelle_uid KEY	str	—	Parcel identifier (year_IDPARCEL).
year KEY	int	year	Harvest year.
scene_date META	date	YYYY-MM-DD	Acquisition date of the Sentinel-2 scene.
NDVI FEATURE	float	[-1, 1]	NDVI value averaged over the parcel polygon for this scene.
EVI FEATURE	float	[-1, 1]*	EVI value averaged over the parcel polygon.
NDWI FEATURE	float	[-1, 1]	NDWI value averaged over the parcel polygon.

Raw files NaN structure: raw files contain ~92% NaN rows. This is **not missing data** — GEE produces one row per (parcel × scene_date) combination, but each Sentinel-2 scene only covers part of the study area. Parcels outside the scene footprint receive NaN. Use `df.dropna(subset=['NDVI'])` to filter to real measurements (~80,000 valid per year, median 42 scenes per parcel per season).

5. Vegetation index formulas

Index	Formula	Meaning
NDVI	$(B8 - B4) / (B8 + B4)$	Normalized Difference Vegetation Index — standard chlorophyll/vigour proxy.
EVI	$2.5 \times (B8 - B4) / (B8 + 6 \times B4 - 7.5 \times B2 + 1)$	Enhanced Vegetation Index — atmosphere/soil-corrected, better for dense canopies.
NDWI	$(B8 - B11) / (B8 + B11)$	Normalized Difference Water Index — sensitive to canopy water content.

Sentinel-2 bands used: B2 (Blue, 490 nm), B4 (Red, 665 nm), B8 (NIR, 842 nm), B11 (SWIR, 1610 nm).

6. Aggregation methodology

- **Cloud masking:** Scene Classification Layer (SCL) — classes 3 (cloud shadow), 7-9 (cloud probabilities), 10 (cirrus), 11 (snow) are masked before computing indices.
- **Scene-level pre-filter:** only scenes with `CLOUDY_PIXEL_PERCENTAGE < 60%` are considered.
- **Per-parcel aggregation:** for each scene, the mean of all valid pixels under the parcel polygon is computed (`reduceRegions`, `scale=20m`).
- **Composite windows (15j, 7j):** for each fixed-length window, the 5 statistics (median, mean, max, min, stdDev) are computed across all valid scenes in the window.
- **Empty windows:** if no valid scene falls in a window (very cloudy periods), the composite returns 15 NaN values for that (parcel × window).

7. Known data quality notes

- **15-day composites:** ~25,8 % NaN rows, concentrated in winter (Dec–Feb).
- **7-day composites:** ~30-35 % NaN expected (smaller windows = more empty periods).
- **EVI outliers:** ~0,12 % of valid rows have $|EVI| > 1,5$ (numerical artifacts). Filter before modelling.
- **NDVI/NDWI correlation:** $r = 0,93$ — near redundant. Consider dimension reduction for non-tree models.

8. Project usage

- **Join key:** `parcelle_uid` matches the parcel sample, weather, soil and target datasets.

- **Suggested derived features:** peak_doy, peak_ndvi, senescence_doy (NDVI < 0,4), senescence_slope, ndvi_integral_AMJ. See [EDA_Report_Sentinel2.pdf](#) for complete recommendations.
- **Recommended granularity per model:** 15-day for RF/XGBoost (fast, low NaN), 7-day for LSTM/GRU (more timesteps), raw for Transformer (maximum sequence length).